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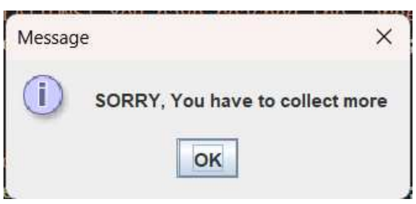
BSCS 2A

MACHINE PROBLEM - OOP

1.

```
import javax.swing.JOptionPane;
public class MP_1 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      int amount = 0;

      input = JOptionPane.showInputDialog("Enter
the amount you have collected: ");
      amount = Integer.parseInt(input);

      if (amount >= 20000) {

         String msg = "CONGRATULATIONS! You
have reached the target amount";
         JOptionPane.showMessageDialog(null,
msg);

      }
      else {
         String msg = "SORRY, You have to collect
more";
         JOptionPane.showMessageDialog(null,
msg);
      }

   }
}
```

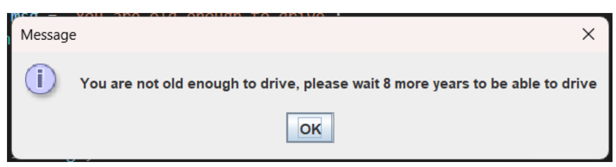
Output:



2.

```
import javax.swing.JOptionPane;
public class MP_2 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      int age = 0, wait = 0;

      input = JOptionPane.showInputDialog("Enter
your age: ");
      age = Integer.parseInt(input);

      if (age >= 18) {

         String msg = "You are old enough to drive";
         JOptionPane.showMessageDialog(null,
msg);

      }
      else {

         wait = 18 - age;
         String msg = "You are not old enough to
drive, please wait " + wait +  " more years to be
able to drive";
         JOptionPane.showMessageDialog(null,
msg);
      }

   }
}
```

Output:



3.

```
import javax.swing.JOptionPane;
public class MP_3 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        double amount = 0, total = 0;

        input = JOptionPane.showInputDialog("Enter
the amount purchased: ");
        amount = Double.parseDouble(input);

        if (amount >= 5000) {

            total = amount - (amount * 0.03);
            String msg = "Please pay: " +
String.format("%.0f",total);
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            String msg = "Please pay: " +
String.format("%.0f",amount);
            JOptionPane.showMessageDialog(null,
msg);
        }

    }
}
```

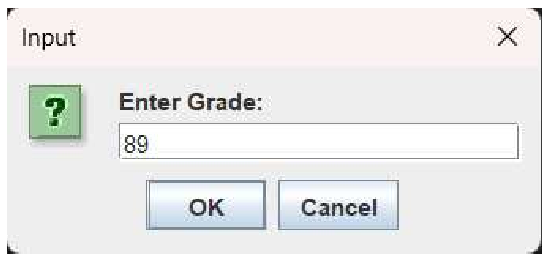
Input:

Output:

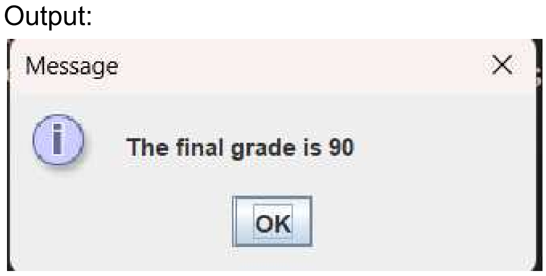
4.

```
import javax.swing.JOptionPane;
public class MP_4 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int grade = 0, fgrade = 0;

        input = JOptionPane.showInputDialog("Enter
Grade: ");
        grade = Integer.parseInt(input);

        if (grade >= 100) {

            String msg = "The grade final is " + grade;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            fgrade = grade + 1;
            String msg = "The final grade is " + fgrade;
            JOptionPane.showMessageDialog(null,
msg);
        }

    }
}
```

Input:



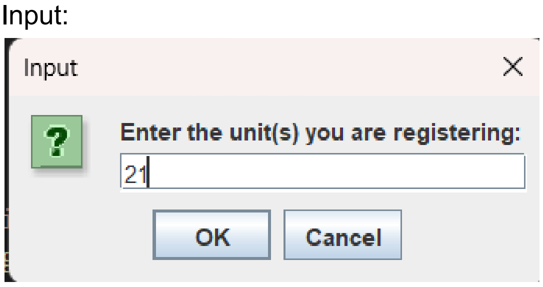
Output:



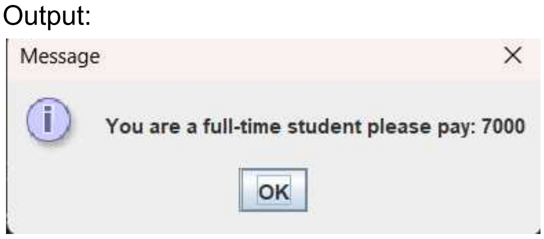
5.

```
import javax.swing.JOptionPane;
public class MP_5 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int units = 0, total = 0;

        input = JOptionPane.showInputDialog("Enter
the unit(s) you are registering: ");
        units = Integer.parseInt(input);

        if (units >= 21) {

            String msg = "You are a full-time student
please pay: 7000";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            total = units * 300;
            String msg = "Your tuition fee is: " + total;
            JOptionPane.showMessageDialog(null,
msg);
        }

    }
}
```

Input:



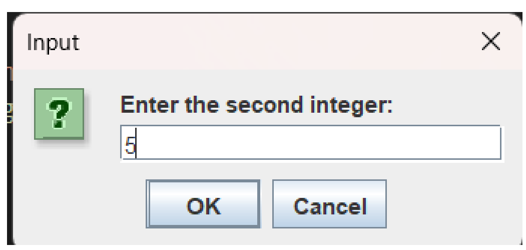
Output:



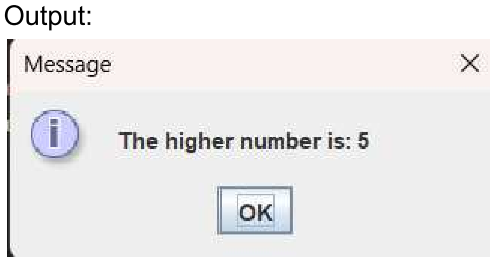
6.

```
import javax.swing.JOptionPane;
public class MP_6 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int fnum = 0, snum = 0;

        input = JOptionPane.showInputDialog("Enter
the first integer: ");
        fnum = Integer.parseInt(input);

        input = JOptionPane.showInputDialog("Enter
the second integer: ");
        snum = Integer.parseInt(input);

        if (fnum > snum) {

            String msg = "The higher number is: " +
fnum;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (snum > fnum){

            String msg = "The higher number is: " +
snum;
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:



Output:



7.

```
import javax.swing.JOptionPane;
public class MP_7 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      int hoursworked = 0, rph = 0;
      double gpay = 0, otpay = 0;

      input = JOptionPane.showInputDialog("Enter
the hours worked: ");
      hoursworked = Integer.parseInt(input);

      input = JOptionPane.showInputDialog("Enter
the rate per hour: ");
      rph = Integer.parseInt(input);

      if (hoursworked <= 40) {

         gpay = hoursworked * rph;
         String msg = "Gross pay = " +
String.format("%.2f",gpay);
         JOptionPane.showMessageDialog(null,
msg);

      }
      else {

         gpay = 40 * rph;
         otpay = (hoursworked - 40) * 1.5 * rph;
         String msg = "Gross pay: " +
String.format("%.2f",gpay) +  " , Overtime pay: " +
String.format("%.2f",otpay);
         JOptionPane.showMessageDialog(null,
msg);
      }

   }
}
```

Output:

8.

```
import javax.swing.JOptionPane;
public class MP_8 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      int speed = 0;

      input = JOptionPane.showInputDialog("Enter
the speed in kilometer per hour (km/h):");
      speed = Integer.parseInt(input);

      if (speed >= 1100) {

         String msg = "It's a civilian aircraft!";
         JOptionPane.showMessageDialog(null,
msg);

      }
      else if (speed >=500) {

         String msg = "It's a military aircraft!";
         JOptionPane.showMessageDialog(null,
msg);
      }
      else {

         String msg = "It's a BIRD!";
         JOptionPane.showMessageDialog(null,
msg);
      }
   }
}
```

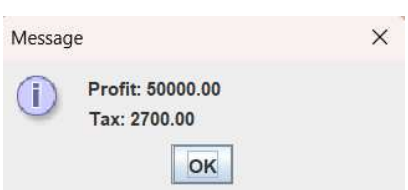
Input:1100

Output:

9.

```
import javax.swing.JOptionPane;
public class MP_9 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        double profit = 0, tax = 0;

        input = JOptionPane.showInputDialog("Enter
sales expenses for the past year: ");
        profit = Double.parseDouble(input);

        if (profit <= 10000) {

            String msg = "Profit: " +
String.format("%.2f",profit) + "\nTax: 0";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (profit <= 30000) {

            tax = 100 + (0.03 * profit);
            String msg = "Profit: " +
String.format("%.2f",profit) + "\nTax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (profit <= 50000) {

            tax = 200 + (0.05 * profit);
            String msg = "Profit: " +
String.format("%.2f",profit) + "\nTax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            tax = 300 + (0.07 * profit);
            String msg = "Profit: " +
String.format("%.2f",profit) + "\nTax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

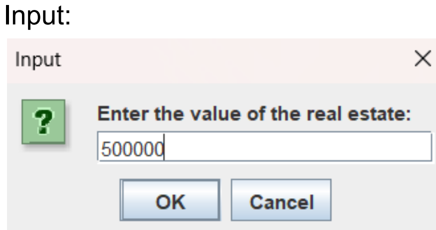
Output:



10.

```
import javax.swing.JOptionPane;
public class MP_10 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        double value = 0, tax = 0;

        input = JOptionPane.showInputDialog("Enter
the value of the real estate: ");
        value = Double.parseDouble(input);

        if (value < 250001.00) {

            tax = value * 0.05;
            String msg = "Real estate tax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (value <= 500000 ) {

            tax = value * 0.1;
            String msg = "Real estate tax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            tax = value * 0.15;
            String msg = "Real estate tax: " +
String.format("%.2f",tax);
            JOptionPane.showMessageDialog(null,
msg);
        }
}
}
```

Input:



Output:

11.

```
import javax.swing.JOptionPane;
public class MP_11 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      double income = 0, tax = 0;
      input = JOptionPane.showInputDialog("Enter
the Income: ");
      income = Double.parseDouble(input);

      if (income <= 5000) {
         String msg = "The tax rate is: 0";
      JOptionPane.showMessageDialog(null, msg);

      }
      else if (income <= 10000) {
         tax = 100 + (0.03 * income);
         String msg = "The tax rate is: " +
String.format("%.2f",tax);
         JOptionPane.showMessageDialog(null,
msg);
      }
      else if (income <= 25000) {
         tax = 200 + (0.06 * income);
         String msg = "The tax rate is: " +
String.format("%.2f",tax);
         JOptionPane.showMessageDialog(null,
msg);
      }
      else if (income <= 50000) {
            tax = 300 + (0.09 * income);
            String msg = "The tax rate is: " +
String.format("%.2f",tax);
         JOptionPane.showMessageDialog(null,
msg);
      }
      else {
         tax = 500 + (0.15 * income);
         String msg = "The tax rate is: " +
String.format("%.2f",tax);
         JOptionPane.showMessageDialog(null,
msg);
      }
   }
}
```

Input: 5000

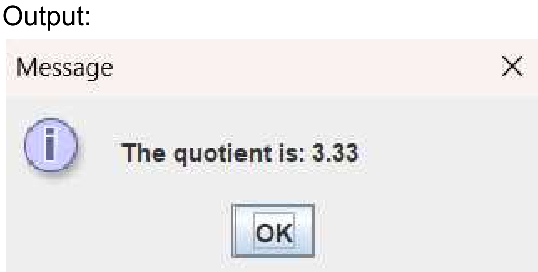
Output:

12.

```
import javax.swing.JOptionPane;
public class MP_12 {
   public static void main(String[] args) throws
Exception {

      String input = " ";
      int choice = 0;
      double result = 0, fnum = 0, snum = 0;

      input = JOptionPane.showInputDialog("1. ADD
TWO NUMBERS \n" +
                              "2. SUBTRACT TWO
NUMBERS \n" +
                              "3. MULTIPLY TWO
NUMBERS \n" +
                              "4. DIVIDE TWO
NUMBERS \n" +
                              "5. EXIT \n" +
                              "\nENTER YOUR
CHOICE:");
      choice = Integer.parseInt(input);

      if (choice == 1) {

         input =
JOptionPane.showInputDialog("Enter the first
number: ");
         fnum = Double.parseDouble(input);

         input =
JOptionPane.showInputDialog("Enter the second
number: ");
         snum = Double.parseDouble(input);

         result = fnum + snum;
         String msg = "The sum is: " +
String.format("%.0f",result);
         JOptionPane.showMessageDialog(null,
msg);

      }
      else if (choice == 2) {

         input =
JOptionPane.showInputDialog("Enter the first
number: ");
         fnum = Double.parseDouble(input);

         input =
JOptionPane.showInputDialog("Enter the second
number: ");
         snum = Double.parseDouble(input);

         result = fnum - snum;
         String msg = "The difference is: " +
String.format("%.0f",result);
         JOptionPane.showMessageDialog(null,
msg);
      }
      else if (choice == 3) {
```

```
        input =
JOptionPane.showInputDialog("Enter the first
number: ");
        fnum = Double.parseDouble(input);

        input =
JOptionPane.showInputDialog("Enter the second
number: ");
        snum = Double.parseDouble(input);

        result = fnum * snum;
        String msg = "The product is: " +
String.format("%.0f",result);
        JOptionPane.showMessageDialog(null,
msg);
      }
      else if (choice == 4) {

        input =
JOptionPane.showInputDialog("Enter the first
number: ");
        fnum = Double.parseDouble(input);

        input =
JOptionPane.showInputDialog("Enter the second
number: ");
        snum = Double.parseDouble(input);

        result = fnum / snum;
        String msg = "The quotient is: " +
String.format("%.2f",result);
        JOptionPane.showMessageDialog(null,
msg);
      }
      else if (choice == 5) {
        JOptionPane.showMessageDialog(null,
"Exiting the program.");
        System.exit(0);
      }
    }
}
```

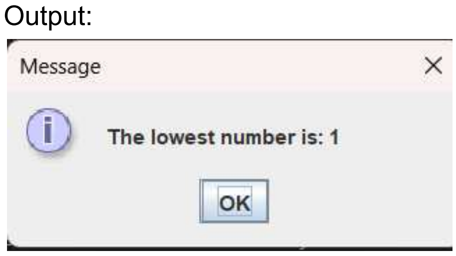
Output:



13.

```
import javax.swing.JOptionPane;
public class MP_13 {
  public static void main(String[] args) throws
Exception {

      String input = " ";
      int num1 = 0, num2 = 0, num3 = 0;

      input = JOptionPane.showInputDialog("Enter
the First Number: ");
      num1 = Integer.parseInt(input);

      input = JOptionPane.showInputDialog("Enter
the Second Number: ");
      num2 = Integer.parseInt(input);

      input = JOptionPane.showInputDialog("Enter
the Third Number: ");
      num3 = Integer.parseInt(input);

      if (num1 < num2 && num1 < num3) {

        String msg = "The lowest number is: " +
num1;
        JOptionPane.showMessageDialog(null,
msg);

      }
      else if (num2 < num1 && num2 < num3) {

        String msg = "The lowest number is: " +
num2;
        JOptionPane.showMessageDialog(null,
msg);
      }
      else {

        String msg = "The lowest number is: " +
num3;
        JOptionPane.showMessageDialog(null,
msg);
      }
    }
}
```

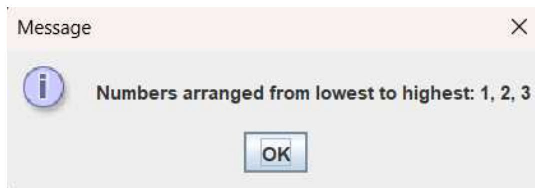
Output:



14.

```
import javax.swing.JOptionPane;
public class MP_14 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int num1 = 0, num2 = 0, num3 = 0;

        input = JOptionPane.showInputDialog("Enter
the First Number: ");
        num1 = Integer.parseInt(input);

        input = JOptionPane.showInputDialog("Enter
the Second Number: ");
        num2 = Integer.parseInt(input);

        input = JOptionPane.showInputDialog("Enter
the Third Number: ");
        num3 = Integer.parseInt(input);

        if (num1 < num2 && num1 < num3 && num2 <
num3) {

            String msg = "Numbers arranged from
lowest to highest: " + num1 + ", " + num2 + ", " +
num3;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num1 < num3 && num1 < num2 &&
num3 < num2) {

            String msg = "Numbers arranged from
lowest to highest: " + num1 + ", " + num3 + ", " +
num2;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num2 < num1 && num2 < num3 &&
num1 < num3) {

            String msg = "Numbers arranged from
lowest to highest: " + num2 + ", " + num1 + ", " +
num3;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num2 < num3 && num2 < num1 &&
num3 < num1) {

            String msg = "Numbers arranged from
lowest to highest: " + num2 + ", " + num3 + ", " +
num1;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num3 < num1 && num3 < num2 &&
num1 < num2) {

            String msg = "Numbers arranged from
lowest to highes: " + num3 + ", " + num1 + ", " +
num2;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            String msg = "Numbers arranged from
lowest to highes: " + num3 + ", " + num2 + ", " +
num1;
            JOptionPane.showMessageDialog(null,
msg);

        }
    }
}
```

Ouput:

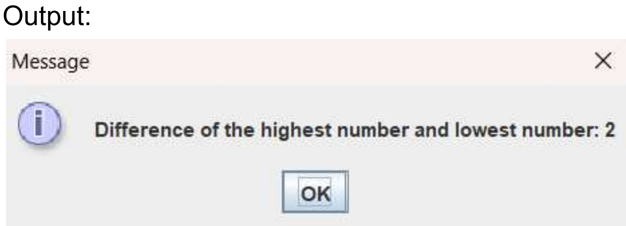


15.

```
import javax.swing.JOptionPane;
public class MP_15 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int num1 = 0, num2 = 0, num3 = 0, diff = 0;

        input = JOptionPane.showInputDialog("Enter
the First Number: ");
        num1 = Integer.parseInt(input);

        input = JOptionPane.showInputDialog("Enter
the Second Number: ");
        num2 = Integer.parseInt(input);

        input = JOptionPane.showInputDialog("Enter
the Third Number: ");
        num3 = Integer.parseInt(input);

        if (num1 < num2 && num1 < num3 && num2 <
num3) {

            diff = num3 - num1;
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num1 < num3 && num1 < num2 &&
num3 < num2) {

            diff = num2 - num1;
```

```
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num2 < num1 && num2 < num3 &&
num1 < num3) {

            diff = num3 - num2;
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num2 < num3 && num2 < num1 &&
num3 < num1) {

            diff = num1 - num2;
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (num3 < num1 && num3 < num2 &&
num1 < num2) {

            diff = num2 - num3;
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            diff = num1 - num3;
            String msg = "Difference of the highest
number and lowest number: " + diff;
            JOptionPane.showMessageDialog(null,
msg);

        }
    }
}
```

Output:

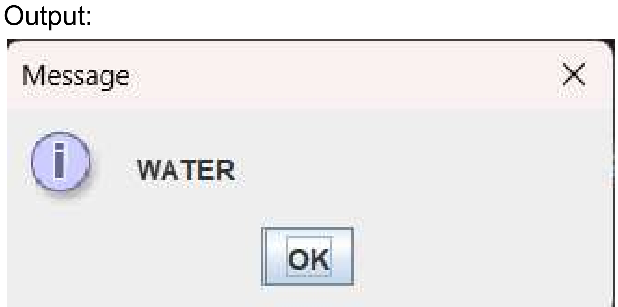


16.

```
import javax.swing.JOptionPane;
public class MP_16 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int TEMP = 0;

        input = JOptionPane.showInputDialog("Enter
the Temperature: ");
        TEMP = Integer.parseInt(input);

        if (TEMP < 0) {

            String msg = "ICE";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (TEMP <= 100) {

            String msg = "WATER";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            String msg = "STEAM";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:

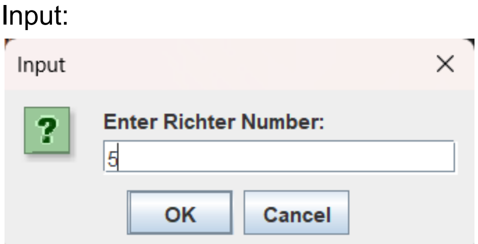
Output:



17.

```
import javax.swing.JOptionPane;
public class MP_17 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        float r = 0;

        input = JOptionPane.showInputDialog("Enter
Richter Number: ");
        r = Float.parseFloat(input);

        if (r < 5) {

            String msg = "LITTLE OR NO DAMAGE";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (r <= 5.5) {

            String msg = "SOME DAMAGE";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (r <= 6.5) {

            String msg = "SERIOUS DAMAGE";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (r <= 7.5) {

            String msg = "DISASTER";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else
        {

            String msg = "CATASTROPHE";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:

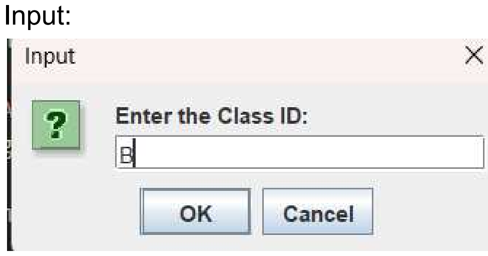


Output:

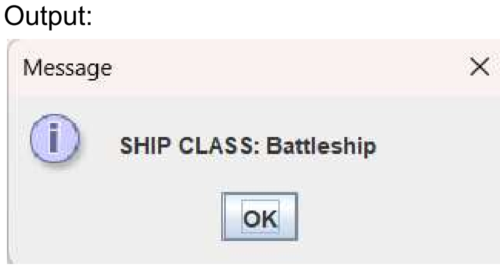
18.

```
import javax.swing.JOptionPane;
public class MP_18 {
    public static void main(String[] args) throws
Exception {

        String classid = " ";

        classid =
JOptionPane.showInputDialog("Enter the Class ID:
");

        if (classid.equalsIgnoreCase("B")) {

            String msg = "SHIP CLASS: Battleship";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (classid.equalsIgnoreCase("C")) {

            String msg = "SHIP CLASS: Cruiser";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (classid.equalsIgnoreCase("D")) {

            String msg = "SHIP CLASS: Destroyer";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (classid.equalsIgnoreCase("F")) {

            String msg = "SHIP CLASS: Frigate";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            String msg = "ERROR CLASS ID";
            JOptionPane.showMessageDialog(null,
msg);
        }
}
}
```

Input:



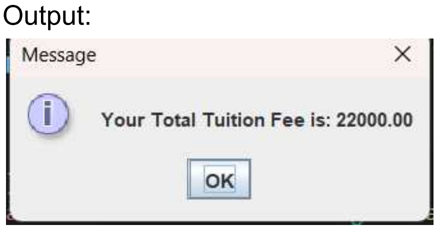
Output:



19.

```
import javax.swing.JOptionPane;
public class MP_19 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int mode = 0;
        double tf = 0, total = 0;

        input = JOptionPane.showInputDialog("Enter
Tuition Fee: ");
        tf = Double.parseDouble(input);

        input = JOptionPane.showInputDialog("(Press
1 for Cash, 2 for Two Installment, 3 for Three
Installment)" +
                                            "\nEnter Mode of
Payment:");
        mode = Integer.parseInt(input);

        if (mode == 1) {

            total = tf - (tf * 0.1);
            String msg = "Your Total Tuition Fee is: " +
String.format("%.2f",total);
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (mode == 2) {

            total = tf - (tf * 0.05);
            String msg = "Your Total Tuition Fee is: " +
String.format("%.2f",total);
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (mode == 3) {
            total = tf + (tf * 0.1);
            String msg = "Your Total Tuition Fee is: " +
String.format("%.2f",total);
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            String msg = "INVALID MODE OF
PAYMENT";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Output:

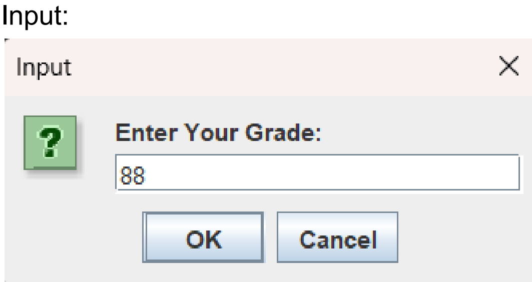


20.

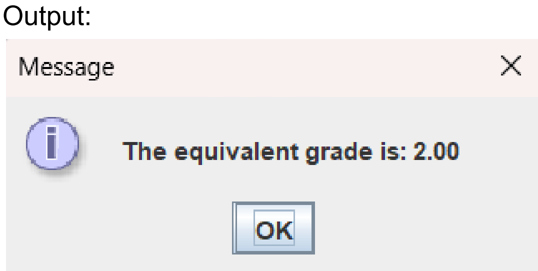
```
import javax.swing.JOptionPane;
public class MP_20 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int grade = 0;

        input = JOptionPane.showInputDialog("Enter
Your Grade: ");
        grade = Integer.parseInt(input);

        if (grade >= 98) {

            String msg = ("The equivalent grade is:
1.00");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (grade >= 95) {

            String msg = ("The equivalent grade is:
1.25");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (grade >= 92) {

            String msg = ("The equivalent grade is:
1.50");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (grade >= 89) {

            String msg = ("The equivalent grade is:
1.75");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 85) {

            String msg = ("The equivalent grade is:
2.00");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 82) {

            String msg = ("The equivalent grade is:
2.25");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 80) {

            String msg = ("The equivalent grade is:
2.50");
```

```
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 77) {

            String msg = ("The equivalent grade is:
2.75");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 75) {

            String msg = ("The equivalent grade is:
3.00");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            String msg = ("OUT OF RANGE");
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:



Output:

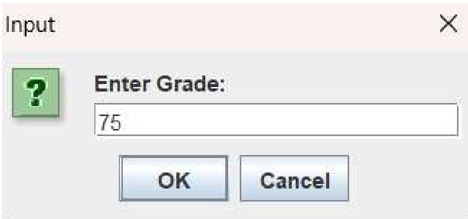


21.

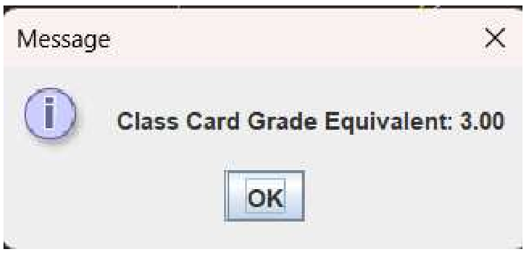
```
import javax.swing.JOptionPane;
public class MP_21 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int grade = 0;

        input = JOptionPane.showInputDialog("Enter
Grade: ");
        grade = Integer.parseInt(input);

        if (grade >= 98) {

            String msg = ("Class Card Grade
Equivalent: 1.00");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 95) {

            String msg = ("Class Card Grade
Equivalent: 1.25");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (grade >= 92) {

            String msg = ("Class Card Grade
Equivalent: 1.50");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (grade >= 89) {

            String msg = ("Class Card Grade
Equivalent: 1.75");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 85) {

            String msg = ("Class Card Grade
Equivalent: 2.00");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 82) {

            String msg = ("Class Card Grade
Equivalent: 2.25");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 80) {

            String msg = ("Class Card Grade
Equivalent: 2.50");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 77) {

            String msg = ("Class Card Grade
Equivalent: 2.75");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 75) {

            String msg = ("Class Card Grade
Equivalent: 3.00");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (grade >= 70) {

            String msg = ("Class Card Grade
Equivalent: 4.00");
```

```
            JOptionPane.showMessageDialog(null,
msg);
        }
        else {

            String msg = ("Class Card Grade
Equivalent: 5.00");
            JOptionPane.showMessageDialog(null,
msg);
    }
}
}
```

Input:



Output:

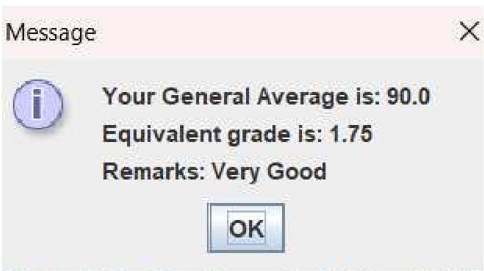


22.

```
import javax.swing.JOptionPane;
public class MP_22 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        double grade = 0, sw = 0, quiz = 0, exam = 0,
ga = 0;

        input = JOptionPane.showInputDialog("Enter
Assignment Grade: ");
        grade = Double.parseDouble(input);

        input = JOptionPane.showInputDialog("Enter
Seatwork Grade: ");
        sw = Double.parseDouble(input);

        input = JOptionPane.showInputDialog("Enter
Quiz Grade: ");
        quiz = Double.parseDouble(input);

        input = JOptionPane.showInputDialog("Enter
Exam Grade: ");
        exam = Double.parseDouble(input);

        ga = (grade * 0.1) + (sw * 0.2) + (quiz * 0.3) +
(exam * 0.4);

        if (ga >= 97) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 1.00" + "\nRemarks:
Excellent");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (ga >= 94) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 1.25" + "\nRemarks:
Excellent");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (ga >= 91) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 1.50" + "\nRemarks:
Very Good");
            JOptionPane.showMessageDialog(null,
msg);

        }
        else if (ga >= 88) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 1.75" + "\nRemarks:
Very Good");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (ga >= 85) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 2.00" + "\nRemarks:
Good");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (ga >= 82) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 2.25" + "\nRemarks:
Good");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (ga >= 79) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 2.50" + "\nRemarks:
Satisfactory");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (ga >= 76) {

            String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 2.75" + "\nRemarks:
Fair");
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (ga == 75) {
```

```
        String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 3.00" + "\nRemarks:
Passed");
        JOptionPane.showMessageDialog(null,
msg);
      }
      else {

        String msg = ("Your General Average is: " +
ga + "\nEquivalent grade is: 5.00" + "\nRemarks:
Failed");
        JOptionPane.showMessageDialog(null,
msg);
    }
}
}
```

Output:



23.

```
import javax.swing.JOptionPane;

public class MP_23 {
    public static void main(String[] args) throws
Exception {

      String input = " ";
      double gpa = 0;

      input = JOptionPane.showInputDialog("Enter
Grade Point Average: ");
      gpa = Double.parseDouble(input);

      if (gpa >= 0.0 && gpa <= 4.0) {

        if (gpa >= 0.0 && gpa <= 0.99) {

          String msg = "GRADE POINT AVERAGE:
" + gpa + "\nTranscript Message: Failed semester -
registration suspended";
          JOptionPane.showMessageDialog(null,
msg);
        } else if (gpa >= 1.0 && gpa <= 1.99) {

          String msg = "GRADE POINT AVERAGE:
" + gpa + "\nTranscript Message: On probation for
next semester";
          JOptionPane.showMessageDialog(null,
msg);
        } else if (gpa >= 2.0 && gpa <= 2.99) {

          String msg = "GRADE POINT AVERAGE:
" + gpa + "\nTranscript Message: (no message)";
          JOptionPane.showMessageDialog(null,
msg);
        } else if (gpa >= 3.0 && gpa <= 3.49) {

          String msg = "GRADE POINT AVERAGE:
" + gpa + "\nTranscript Message: Dean's list for
semester";
          JOptionPane.showMessageDialog(null,
msg);
        } else if (gpa >= 3.5 && gpa <= 4.0) {

          String msg = "GRADE POINT AVERAGE:
" + gpa + "\nTranscript Message: Highest honors
for semester";
          JOptionPane.showMessageDialog(null,
msg);
        }
      } else {

        String msg = "Invalid GPA. Please enter a
GPA within the range 0.0 to 4.0.";
        JOptionPane.showMessageDialog(null,
msg);
      }
    }
}
```

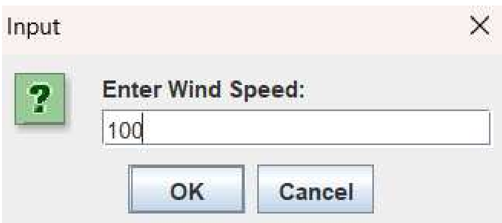
Output:

24.

```
import javax.swing.JOptionPane;
public class MP_24 {
    public static void main(String[] args) throws
Exception {

      String input = " ";
      int speed = 0;

      input = JOptionPane.showInputDialog("Enter
Wind Speed: ");
      speed = Integer.parseInt(input);

      if (speed < 25) {

        String msg = "Not a strong wind";
        JOptionPane.showMessageDialog(null,
msg);
      }
      else if (speed <= 38) {

        String msg = "Strong wind";
        JOptionPane.showMessageDialog(null,
msg);
      }
      else if (speed <= 54) {
```

```
            String msg = "Gale";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else if (speed <= 72) {

            String msg = "Whole Gale";
            JOptionPane.showMessageDialog(null,
msg);
        }
        else
        {
            String msg = "Hurricane";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:

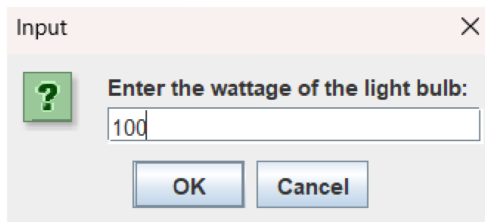


Output:

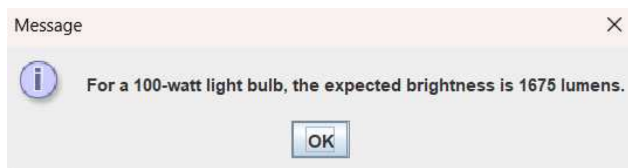
25.

```
import javax.swing.JOptionPane;

public class MP_25 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        int watts = 0;
        int lumens = 0;

        input = JOptionPane.showInputDialog("Enter
the wattage of the light bulb: ");
        watts = Integer.parseInt(input);

        switch (watts) {
            case 15:
                lumens = 125;
                break;
            case 25:
                lumens = 215;
                break;
            case 40:
                lumens = 500;
                break;
            case 60:
                lumens = 880;
                break;
            case 75:
                lumens = 1000;
                break;
            case 100:
                lumens = 1675;
                break;
            default:
                lumens = -1;
                break;
        }

        if (lumens == -1) {
            JOptionPane.showMessageDialog(null,
"Invalid wattage. Please enter a valid wattage.");
        } else {
            String msg = "For a " + watts + "-watt light
bulb, the expected brightness is " + lumens + "
lumens.";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:



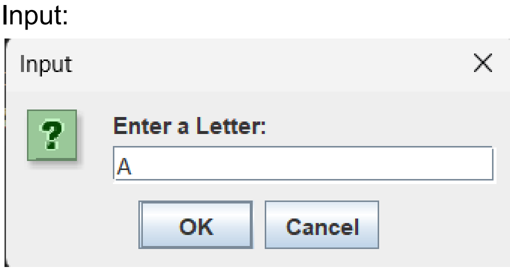
Output:



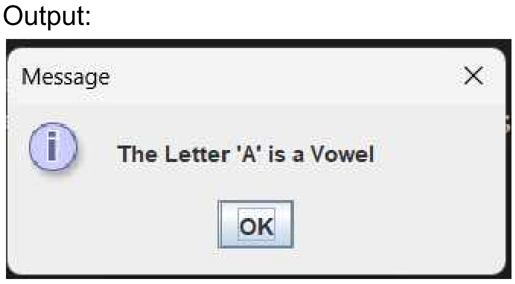
26.

```
import javax.swing.JOptionPane;

public class MP_26 {
    public static void main(String[] args) throws
Exception {

        String letter = " ";

        letter = JOptionPane.showInputDialog("Enter
a Letter: ");

        if (letter.equalsIgnoreCase("A") ||
letter.equalsIgnoreCase("E") ||
            letter.equalsIgnoreCase("I") ||
letter.equalsIgnoreCase("O") ||
            letter.equalsIgnoreCase("U")) {

            String msg = "The Letter '" + letter + "' is a
Vowel";
            JOptionPane.showMessageDialog(null,
msg);

        }
        else {

            String msg = "The Letter '" + letter + "' is a
Consonant";
            JOptionPane.showMessageDialog(null,
msg);
        }
    }
}
```

Input:



Output:



27.

```
import javax.swing.JOptionPane;

public class MP_27 {
    public static void main(String[] args) {

        String input = " ";
        int V_Class = 0;
        double rate = 0, addfee = 0, km = 0, total;
        input = JOptionPane.showInputDialog("Enter
the vehicle class (1, 2, or 3): ");
        V_Class = Integer.parseInt(input);


        if (V_Class == 1) {

            rate = 41;
            addfee = 20;

        } else if (V_Class == 2) {

            rate = 102;
            addfee = 35;

        } else if (V_Class == 3) {

            rate = 122;
            addfee = 50;

        } else {

            JOptionPane.showMessageDialog(null,
"Error: Invalid vehicle class. Please enter 1, 2, or
3.");
            return;
        }

        input = JOptionPane.showInputDialog("Enter
the distance in kilometers: ");
        km = Double.parseDouble(input);

        total = rate + (km * addfee);

        String msg = String.format("Vehicle Class: " +
V_Class + "\nRate: " + rate + "\nAdditional Fee per
km: " + addfee + "\nTotal Cost: " + total);

        JOptionPane.showMessageDialog(null, msg);
    }
}
```

Output:

28.

```
import javax.swing.JOptionPane;

public class MP_28 {
    public static void main(String[] args) {

        String input = " ";
        String scholarship = " ";
        double totalGrade = 0, grade = 0, average = 0,
tfd = 0;

        for (int i = 1; i <= 10; i++) {

            input =
JOptionPane.showInputDialog("Enter grade for
Subject " + i + ": ");
            grade = Double.parseDouble(input);

            if (grade < 0 || grade > 100) {
                JOptionPane.showMessageDialog(null,
"Error: Invalid grade. Please enter a valid grade
between 0 and 100.");
                return;
            }

            totalGrade += grade;
        }

        average = totalGrade / 10;

        if (average >= 98) {

            scholarship = "President's List";
            tfd = 100;
        } else if (average >= 95) {

            scholarship = "College Scholar";
            tfd = 50;
        } else if (average >= 92) {

            scholarship = "Dean's List";
            tfd = 30;
        } else {

            scholarship = "None";
            tfd = 0;
        }

        String msg = ("Average Grade: " +
String.format("%.2f",average) +
                    "\nScholarship Grant:" + scholarship +
"\nTuition Fee Discount: " + tfd + "%");
        JOptionPane.showMessageDialog(null, msg);
    }
}
```

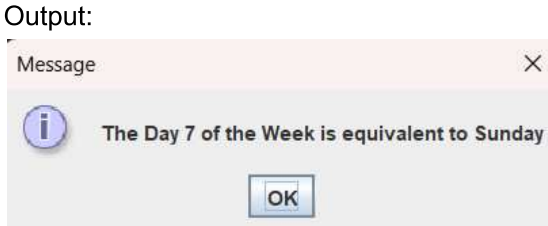
Output:

29.

```
import javax.swing.JOptionPane;

public class MP_29 {
    public static void main(String[] args) throws
Exception {

        String input = " ";
        String eq = " ";
        int NOWeek = 0;

        input = JOptionPane.showInputDialog("Enter
Day of Week: ");
        NOWeek = Integer.parseInt(input);

        switch (NOWeek) {
            case 1:
                eq = "Monday";
                break;
            case 2:
                eq = "Tuesday";
                break;
            case 3:
                eq = "Wednesday";
                break;
            case 4:
                eq = "Thursday";
                break;
            case 5:
                eq = "Friday";
                break;
            case 6:
                eq = "Saturday";
                break;
            case 7:
                eq = "Sunday";
                break;
            default:

            JOptionPane.showMessageDialog(null,
"Error: Invalid day of the week. Please enter a
value between 1 and 7.");
            return;
        }

            String msg = "The Day " + NOWeek + " of
the Week is equivalent to " + eq;
            JOptionPane.showMessageDialog(null,
msg);
    }
}
```

Input: 7

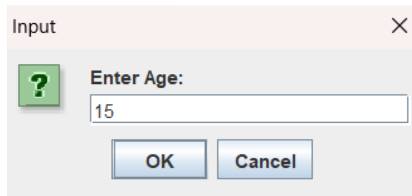
Output:



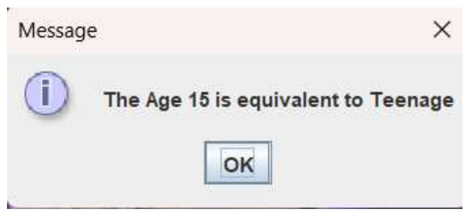
30.

```
import javax.swing.JOptionPane;

public class MP_30 {
    public static void main(String[] args) {

        String input = " ";
        String category = " ";
        int age = 0;

        input = JOptionPane.showInputDialog("Enter
Age: ");
        age = Integer.parseInt(input);

        switch (age) {
            case 60:
            case 59:
            case 58:
            case 57:
            case 56:
            case 55:
            case 54:
            case 53:
            case 52:
            case 51:
            case 50:
                category = "Adults";
                break;
            case 40:
            case 39:
            case 38:
            case 37:
            case 36:
            case 35:
            case 34:
            case 33:
            case 32:
            case 31:
            case 30:
                category = "Youth";
                break;
            case 19:
            case 18:
            case 17:
            case 16:
            case 15:
            case 14:
            case 13:
            case 12:
            case 11:
            case 10:
                category = "Teenage";
                break;
            case 9:
            case 8:
            case 7:
            case 6:
            case 5:
            case 4:
            case 3:
            case 2:
            case 1:
            case 0:
                category = "Kids";
                break;
            default:
                JOptionPane.showMessageDialog(null,
"The Age " + age + " is equivalent to Senior
Citizen");
                return;
        }

        String msg = "The Age " + age + " is
equivalent to " + category;
        JOptionPane.showMessageDialog(null, msg);
    }
}
```

Input:



Output:

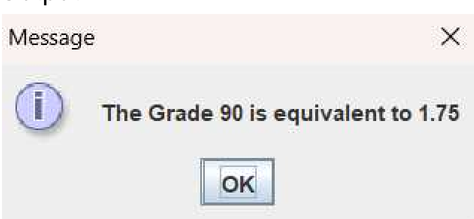


31.

```
import javax.swing.JOptionPane;

public class MP_31 {
    public static void main(String[] args) {

        int grade;
        String input, equivalent = " ";

        input = JOptionPane.showInputDialog("Enter
Grade: ");
        grade = Integer.parseInt(input);

        switch (grade) {
            case 100:
            case 99:
            case 98:
                equivalent = "1.0";
                break;
            case 97:
            case 96:
            case 95:
                equivalent = "1.25";
                break;
            case 94:
            case 93:
            case 92:
                equivalent = "1.5";
                break;
            case 91:
            case 90:
            case 89:
                equivalent = "1.75";
                break;
```

```
            case 88:
            case 87:
            case 86:
                equivalent = "2.0";
                break;
            case 85:
            case 84:
            case 83:
                equivalent = "2.25";
                break;
            case 82:
            case 81:
            case 80:
                equivalent = "2.5";
                break;
            case 79:
            case 78:
            case 77:
                equivalent = "2.75";
                break;
            case 76:
            case 75:
                equivalent = "3.0";
                break;
            default:
                equivalent = "5.0";
                break;
        }

        String msg = "The Grade " + grade + " is
equivalent to " + equivalent;
        JOptionPane.showMessageDialog(null, msg);
    }
}
```

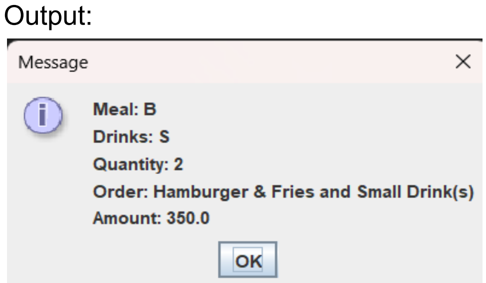
Output:



32.

```
import javax.swing.JOptionPane;

public class MP_32 {
    public static void main(String[] args) {

        String input = " ";
        String meal = " ", drink = " ", dName = " ",
mName = " ";
        int quantity = 0;
        double mealPrice = 0, drinkPrice = 0,
totalAmount = 0;

        meal = JOptionPane.showInputDialog("Enter
Meal (A/B/C/D): ");

        drink = JOptionPane.showInputDialog("Enter
Drinks (S/M/L): ");

        input = JOptionPane.showInputDialog("Enter
Quantity: ");
        quantity = Integer.parseInt(input);

        switch (meal) {
            case "A":
                mName = "Chicken & Spaghetti";
                mealPrice = 150;
                break;
            case "B":
                mName = "Hamburger & Fries";
                mealPrice = 145;
                break;
            case "C":
                mName = "Cheesedog";
                mealPrice = 100;
                break;
            case "D":
                mName = "Pizza";
                mealPrice = 80;
                break;
            default:
                JOptionPane.showMessageDialog(null,
"Invalid Meal Selection");
                return;
        }

        switch (drink) {
            case "S":
                dName = "Small Drink(s)";
                drinkPrice = 30;
                break;
            case "M":
                dName = "Medium Drink(s)";
                drinkPrice = 35;
                break;
            case "L":
                dName = "Large Drink(s)";
                drinkPrice = 40;
                break;
            default:
                JOptionPane.showMessageDialog(null,
"Invalid Drink Selection");
                return;
        }

        totalAmount = (mealPrice + drinkPrice) *
quantity;

        String msg = ("Meal: " + meal + "\nDrinks: " +
drink + "\nQuantity: " + quantity +
                    "\nOrder: " + mName + " and " +
dName + "\nAmount: " + totalAmount);

        JOptionPane.showMessageDialog(null, msg);
    }
}
```

Output:



33.

```
import javax.swing.JOptionPane;

public class MP_33 {
   public static void main(String[] args) {

      String input = " ";
      String room = " ", re = " ", pay = " ";
      int code = 0;
      double rate = 0, meal = 0, amount = 0, total =
0;

      room = JOptionPane.showInputDialog("Enter
Room Type (A/B/C): ");

      input = JOptionPane.showInputDialog("Enter
Code of Payment (1/2): ");
      code = Integer.parseInt(input);

      switch (room) {
         case "A":
            re = "Standard";
            rate = 5000;
            meal = 1500;
            break;
         case "B":
            re = "De Luxe";
            rate = 15000;
            meal = 2000;
            break;
         case "C":
            re = "Suite";
            rate = 20000;
            meal = 3000;
            break;
         default:
            JOptionPane.showMessageDialog(null,
"Invalid Room Type");
            return;
      }

      amount = rate + meal;
      switch (code) {
         case 1:
            pay = "Cash";
            total = (0.02 * amount) - amount;
            break;
         case 2:
            pay = "Credit Card";
            total = (0.05 * amount) + amount;
            break;
         default:
            JOptionPane.showMessageDialog(null,
"Invalid Code");
            return;
      }

      String msg = ("Room Type: " + room +
"\nRoom is: " + re + "\nCode of Payment: " + code
+ " (" + pay + ")\n" +
               "\nAmount: " + String.format("%.2f",
total));
      JOptionPane.showMessageDialog(null, msg);
   }
}
```

Output:

34.

```
import javax.swing.JOptionPane;
public class MP_34 {
   public static void main(String[] args) {

      String input = " ";
      int kwh = 0;
      double tBill = 0;

      input = JOptionPane.showInputDialog("Enter
number of kilowatt hours: ");
      kwh = Integer.parseInt(input);

      if (kwh <= 99) {

         tBill = kwh * 10.00;
      } else if (kwh <= 249) {

         tBill = 99 * 10.00 + (kwh - 99) * 50.00;
      } else {

         tBill = 99 * 10.00 + (249 - 99) * 50.00 + (kwh
- 249) * 100.00;
      }

      String msg = ("Total electric bill is P" +
String.format("%.2f", tBill));
      JOptionPane.showMessageDialog(null, msg);
   }
}
```

Input: 100

Output:

35.

```
import javax.swing.JOptionPane;
public class MP_35 {
    public static void main(String[] args) throws Exception {

        String rps1 = " ", rps2 = " ";

        rps1 = JOptionPane.showInputDialog("Player 1 Enter your choice: ");
        rps2 = JOptionPane.showInputDialog("Player 2 Enter your choice: ");

        if (rps1.equalsIgnoreCase("P") && rps2.equalsIgnoreCase("R")) {

            String msg = "Player 1 Wins! PAPER COVERS ROCK";
            JOptionPane.showMessageDialog(null, msg);

        }
        else if (rps1.equalsIgnoreCase("R") && rps2.equalsIgnoreCase("S")) {

            String msg = "Player 1 Wins! ROCK BREAKS SCISSOR";
            JOptionPane.showMessageDialog(null, msg);
        }
        else if (rps1.equalsIgnoreCase("S") && rps2.equalsIgnoreCase("P")) {

            String msg = "Player 1 Wins! SCISSOR CUTS PAPER";
            JOptionPane.showMessageDialog(null, msg);
        }
        else if (rps1.equalsIgnoreCase("R") && rps2.equalsIgnoreCase("P")) {

            String msg = "Player 2 Wins! PAPER COVERS ROCK";
            JOptionPane.showMessageDialog(null, msg);

        }
        else if (rps1.equalsIgnoreCase("S") && rps2.equalsIgnoreCase("R")) {

            String msg = "Player 2 Wins! ROCK BREAKS SCISSOR";
            JOptionPane.showMessageDialog(null, msg);
        }
        else if (rps1.equalsIgnoreCase("P") && rps2.equalsIgnoreCase("S")) {

            String msg = "Player 2 Wins! SCISSOR CUTS PAPER";
            JOptionPane.showMessageDialog(null, msg);
        }
        else if (rps1.equalsIgnoreCase(rps2)) {

            String msg = "It's A Draw! NOBODY WINS";
            JOptionPane.showMessageDialog(null, msg);
        }
        else {

            String msg = "ERROR INPUT";
            JOptionPane.showMessageDialog(null, msg);
        }
    }
}
```

Input: Player 1 = P, Player 2 = S

Output: